2.4 Unit Tangent, Unit Normal, and Binormal Vectors Worksheet

1. Let $\mathbf{r}(t) = \langle t^2, t^2 \sin(t), t^2 \cos(t) \rangle$.

   a) Give parametric equations for the line tangent to $\mathbf{r}(t)$ at $t = \pi$.

   b) Determine the unit tangent vector to $\mathbf{r}(t)$ at $t = \pi$.

2. Consider the curve defined by $\mathbf{r}(t) = \langle t, 4 - t^2, \ln(t) \rangle$.

   a) Find the point on the curve at which the tangent vector is parallel to the vector $\langle 2, -1, 8 \rangle$.

b) Find parametric equations of the tangent line at the point you found in (a).

3. For the curve given by $\mathbf{r}(t) = \langle 4\sin(t), 3t, 4\cos(t) \rangle$.

a) Find the unit tangent vector $\mathbf{T}(t)$

b) Find the unit normal vector $\mathbf{N}(t)$.

c) Find the binormal vector $\mathbf{B}(t)$.

4. Consider the curve $\mathbf{r}(t)=\left\langle t,t^2,\frac{1}{3}t^3\right\rangle$.

a) Find the unit tangent $\mathbf{T}$ at the point $(0,0,0)$.

b) Find the binormal $\mathbf{B}$ at the point $(0,0,0)$.

c) Determine the equation of the plane which contains **T** and **N**.